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SUMMARY

Full-stack engineer building production systems end-to-end, from **React** and **TypeScript** frontends to **Spring Boot**, **Flask**, and **Node** backends on **AWS**, including the AI layer. Cut page load **35%** with a **GraphQL** gateway, infrastructure cost by **30%** with **Redis** caching, and delivered **300ms on-device LLM inference** via INT8-quantized **ONNX Runtime**. **ICPC Regionalist**.

EXPERIENCE

Software Engineer | **TechMentee, Inc.**, USA **June 2026 – Present**

- Built a configuration-driven **Python** pipeline for an E-Pharma platform that automates ingestion, validation, feature engineering, and model training, replacing ad-hoc analysis scripts with reusable, testable components.
- Designed feature engineering and training workflows with lag/rolling features, multi-model evaluation, and joblib artifact persistence, structured for integration with the **Node.js** backend.

Software Engineer | **Kahana** (AI Agentic Browser Startup), USA **June 2025 – June 2026**

- **Achieved a 98% command success rate** for voice-driven browser actions (open tab, group tabs, search history) built on a **Python** **Flask** backend and a **React** with **TypeScript** frontend with Tailwind CSS, improving cross-device experience.
- **Reduced page load time by 35%** by consolidating separate authentication, recommendation, and search-suggestion calls behind a single **GraphQL** gateway over **Flask** microservices, with **Kafka** events keeping recommendations fresh.
- **Cut infrastructure costs by 30%** with a **Redis** cache-aside layer that repopulates on misses and publishes **Kafka** events to refresh recommendations, plus **RabbitMQ** and **Celery** async jobs.
- **Improved frontend performance** by restructuring React state from arrays to **hash maps** for O(1) lookups, adding **memoization** to cut re-renders, and using code splitting and lazy loading to shrink initial load.
- **Built a two-tier intent engine** that races an **on-device MiniLM** matcher against a **Gemini** path in parallel threads, taking the higher-confidence result and canceling the slower path via an atomic flag to save compute.
- **Optimized on-device inference to a 300ms round trip** by tuning an **ONNX Runtime BERT** pipeline for low-resource devices with multithreading and INT8 quantization, matching intents to a **function registry** of typed browser handlers.

Software Engineer | **Programmers Army**, India **May 2022 – Aug 2023**

- **Reduced p95 latency from 450ms to 220ms** under peak load by building Spring Boot REST services with Spring Data JPA and Hibernate for payments, orders, and assessments, while improving testability through dependency injection and slice tests.
- **Strengthened application security** and reduced authentication-related support tickets by implementing an authentication service with Spring Security, JWT, OAuth2, and role-based access control.
- **Maintained consistent search performance during traffic spikes** across a **40K+ daily active user base** by integrating Elastic-search for full-text search across courses, videos, and PDFs with per-field boosting and fuzzy matching.
- **Achieved 80% code coverage** and improved release quality by implementing automated tests with JUnit and Mockito and integrating SonarQube for static analysis.

PROJECTS

DocQuery [[GitHub](#)] [[Live Demo](#)] | FastAPI, Gradio, ChromaDB, Sentence-Transformers, Groq/Ollama, Docker **Jan 2025 – July 2025**

- Built a **RAG pipeline** that extracts PDF text with **pypdf**, chunks with overlap, embeds locally using **sentence-transformers (all-MiniLM-L6-v2)**, and stores vectors in persistent **ChromaDB** with source, page, and chunk metadata.
- **Eliminated citation hallucination** by grounding every answer in stored vector metadata, so responses cite the exact PDF filename, page, and chunk rather than model-generated references.
- Built a **ReAct agent** with typed tools for question answering, summarization, and structured report generation behind a pluggable LLM interface supporting **Groq Llama 3.3 70B** and local **Ollama llama3.2:3b**.
- Shipped a **Gradio UI** for PDF upload and Q&A plus **FastAPI** endpoints for ingestion, querying, health checks, and agent routing, deployed live on **Hugging Face Spaces**.

High-Performance Search Engine | C++, Node.js (N-API), Express [[GitHub](#)] [[Live Demo](#)] **2025**

- Implemented a **Trie**-backed search engine in **C++** with an **inverted index** supporting prefix and exact lookups, plus recursive **Levenshtein** traversal for fuzzy matching within edit distance 2.
- Reduced Trie node memory by 42% by replacing **unordered_map** child storage with a contiguous vector, validated under AddressSanitizer, and exposed the engine as a native N-API addon with real-time latency and throughput monitoring.

PocketSplit – Expense Sharing App | React Native, Expo, Spring Boot, PostgreSQL, Docker [[GitHub](#)] **Aug 2024 – Jan 2025**

- Built a cross-platform expense-sharing app end-to-end: a **React Native** and **Expo** client for splitting bills, tracking friend and group balances, and syncing contacts, with sessions stored via **Expo SecureStore**.
- Built a **Spring Boot** and **PostgreSQL** backend with token-authenticated **REST APIs**, **Flyway** migrations, and JWT auth, containerized with **Docker Compose** and published to **GitHub Container Registry** via **GitHub Actions**.

TECHNICAL SKILLS

Languages:	Python, TypeScript, Java, C++, SQL
Frontend:	React , Next.js, Angular, React Native, Redux, Tailwind CSS, HTML, CSS
Backend:	Flask, Spring Boot , Django, Node.js, GraphQL, REST APIs, Microservices, Kafka, RabbitMQ, Celery
Databases:	PostgreSQL, MySQL, Redis, Elasticsearch, ChromaDB, pgvector
Infrastructure:	AWS (ECS Fargate, Lambda, S3, RDS, CloudFront, API Gateway), Azure, Docker, Kubernetes, GitHub Actions
AI/ML:	RAG, Gemini API, sentence-transformers, MiniLM, embeddings, ONNX Runtime, scikit-learn, Ollama
Testing:	pytest, JUnit, Mockito, Jest, SonarQube

EDUCATION

Indiana University Bloomington | Bloomington, IN, USA **Aug 2023 – May 2025**

Master of Science in Data Science

Courses: Applied Algorithms, Software Engineering, Advanced Database Concepts, Computer Vision

University of Mumbai | Mumbai, India **Aug 2018 – May 2022**

Bachelor of Engineering in Information Technology

Courses: Object-Oriented Programming, Computer Networks, Operating Systems, Cryptography & Network Security